

Scope Statement

Natural Math models growth processes that share four properties:

1. Tree-like or forest-like structure. The parent relation forms a directed forest with no cycles. What grows is a branching hierarchy, not a general network.
2. Local decision-making. Each growing tip acts on information within a finite neighborhood. No global coordination, no central planner, no non-local state.
3. Finite resource budget. Growth, sensing, and retreat all cost energy. When energy is exhausted, activity stops. This is a theorem, not an assumption.
4. Spatial embedding. Positions matter. Growth happens in three-dimensional integer space with exact rational geometry.

This covers plant roots and shoots, fungal mycelium, vascular networks, neural arborization, and certain infrastructure systems. It does not cover cyclic networks, systems requiring global equilibrium, non-spatial growth, or independent reproduction. Those require different frameworks.

Reading Guide

If you are... Read... To understand...
New to the framework Part 1: Intuitive Overview What it does and why, no math
Implementing the system Part 2: Formal Specification Every definition, rule, and invariant
Curious about design choices Part 3: Design Rationale Why synchronous? Why SENSE costs energy?
Trying to apply it Part 4: Worked Examples Step-by-step in specific domains

Part 1: Intuitive Overview

What Natural Math Models

Natural Math describes how branching structures grow, adapt, and halt—using only local information and a finite energy budget. Think of roots navigating soil, fungi exploring a forest floor, or neurons finding targets. No single tip sees the whole picture. Each makes simple choices based on what's nearby. The large-scale shape—the branching pattern, the zones of dense exploration, the points of retreat—emerges from thousands of local decisions.

The Core Idea

At each moment, a growing tip faces one of four choices:

- EXTEND: Keep growing forward, or split into two branches. Costs energy, explores new territory.
- SENSE: Wait, adjust direction, probe the boundary. Costs less energy but doesn't move.
- RESTRICT: Stop permanently. Energy too low, or neighbor too close. This branch is done.
- CONSERVE: Pull back and return unused energy to where you came from. The parent can reallocate it.

These four choices, made locally and repeatedly, generate the full range of behaviors: growth, branching, adaptation, retraction, and halt.

The Site Lifecycle

Every site in the system moves through states in a predictable order. Each state has both an entry condition and an exit condition—no site is ever stuck without a path forward (even if that path is death).

State	Decision	Entry Condition	Exit Condition
Growth	EXTEND	Energy available, space ahead, gradient present	Low energy, proximity blocked, pressure-triggered retreat
Exploration	SENSE	Flat gradient, near a boundary, or waiting	Gradient appears, energy runs low, or conditions trigger retreat
Retreat	CONSERVE	High pressure or external hibernate signal	N/A — site is removed; energy returns to parent
Death	RESTRICT	Energy below threshold, or neighbor too close	None — site is permanently frozen

This lifecycle is complete for an individual growth structure. It captures the full arc from seed to final state. What it does not capture—yet—is what happens when a structure is so successful that local growth is no longer the best use of its resources. That extension (reproduction) is specified in Open Question #8.

How Tips Decide

A tip senses only what's nearby—other tips within range, obstacles it contacts directly, and a memory trail left by previous growth. It compares energy levels (moving toward richer areas), avoids places already visited, and responds to accumulated pressure (frustration at obstacles that eventually triggers branching or breakthrough).

The Three Strategies at an Obstacle

When a tip hits an obstacle:

- Dither (SENSE): Wait and probe. Cheap (2 energy per step) but doesn't move. At a permanent obstacle, the tip eventually runs out of energy and stops.
- Push through (repeated EXTEND): Hit the obstacle repeatedly. Expensive (10 energy per hit) but pressure builds up. At a threshold, the tip either bifurcates around the obstacle or erodes through it.
- Retreat (CONSERVE): Give up on this direction, return leftover energy to the parent. Cheap (1 energy plus recovery loss) and preserves resources for other branches.

No central planner chooses. The tip's local energy and pressure determine the outcome.

Energy is Everything

Every action costs energy. The system starts with a fixed total. When it's gone, growth stops. Even CONSERVE loses a fraction of energy (no perfect recycling), so the system cannot oscillate forever. Closed systems always reach an inactive state. This is a mathematical guarantee.

If you add an external energy source—sunlight, rainfall, investment—the system can persist. The framework handles both. The theorem tells you exactly when halting is guaranteed.

What Emerges

From these simple rules, across domains, you get:

- Branching networks that efficiently explore space
- Retraction and reallocation when resources are scarce
- Pressure-driven breakthroughs at obstacles
- Trail-based memory that avoids redundant exploration
- Global patterns from purely local decisions

The framework does not design the shape. It grows it.

Part 2: Formal Specification

1. The Lattice

1.1 Definition

The ambient space is the integer lattice:

$$\Lambda = \mathbb{Z}^3$$

This is locally finite: any ball of finite radius contains finitely many points.

1.2 Position Function

Each site p has a unique position:

$$\text{pos}: \mathcal{A}_t \rightarrow \Lambda$$

Positions are not required to be unique across sites, but conflict resolution rules (Section 8) prevent two sites from occupying the same position after an update.

2. State Definition

2.1 Global State

$$S_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P})$$

where:

- $\mathcal{A}_t$ is a finite set of site identifiers.
- $\mathcal{E}_t : \mathcal{A}_t \rightarrow \mathbb{Q}_{\geq 0}$ assigns energy to each active site.
- $\mathcal{P}$ is the fixed parameter tuple (all values in $\mathbb{Q}_{\geq 0}$ unless noted).

2.2 Site State

Each site $p \in \mathcal{A}_t$ carries:

$$x_{p,t} = (\text{pos}_p, \text{dir}_p, \text{type}_p, \text{pressure}_p, \text{parent}_p)$$

Field Type Description

$\text{pos}_p \in \mathbb{Z}^3$	Position in lattice
$\text{dir}_p \in \mathbb{Z}^3 \setminus \{0\}$	Direction vector
$\text{type}_p \in \{\text{seed}, \text{tip}, \text{branch}, \text{sensor}, \text{inert}\}$	Behavioral type
$\text{pressure}_p \in \mathbb{Q}_{\geq 0}$	Accumulated constraint
$\text{parent}_p \in \mathcal{A}_t \cup \{\text{nothing}\}$	Parent site identifier

2.3 Total Energy

$$E_{\{\text{tot}\}}(t) = \sum_{p \in \mathcal{A}_t} E(p,t)$$

Energy lost to dissipation is not tracked separately; it is the difference $E_{\{\text{tot}\}}(0) - E_{\{\text{tot}\}}(t)$.

2.4 System Invariants

The following hold for all $t \geq 0$:

I1 (Parent acyclicity): The parent relation forms a directed forest with edges from child to parent. For all $p \in \mathcal{A}_t$, if $\text{parent}_p \neq \text{varnothing}$, then parent_p was created at a strictly earlier timestep than p . Therefore there are no cycles.

I2 (Inert energy): For all $p \in \mathcal{A}_t$ with $\text{type}_p = \text{inert}$, $E(p,t) = 0$.

I3 (RESTRICT energy bound): For all $p \in \mathcal{A}_t$ that chose RESTRICT at $t-1$, either $E(p,t) < \tau$ (energy trigger) or $\min Q < \iota^2$ was true (proximity trigger). In both cases, RESTRICT sets $E(p,t) = 0$ and $\text{type}_p = \text{inert}$, satisfying I2.

I4 (Active sites only): Only sites with $\text{type}_p \in \{\text{seed}, \text{tip}, \text{branch}, \text{sensor}\}$ are evaluated by the decision function. Inert sites are never evaluated.

I5 (Finite support): $|\mathcal{A}_t|$ is finite for all t . The trail map has finite support at all t .

3. Neighborhood

3.1 Quadrance

For positions $u, v \in \mathbb{Z}^3$:

$$Q(u, v) = (u_x - v_x)^2 + (u_y - v_y)^2 + (u_z - v_z)^2 \in \mathbb{Z}_{\geq 0}$$

3.2 Neighborhood Definition

For site $p \in \mathcal{A}_t$ with integer quadrance bound $R \in \mathbb{Z}_{>0}$:

$$\mathcal{N}_R(p, t) = \{ q \in \mathcal{A}_t \setminus \{p\} : Q(\text{pos}_p, \text{pos}_q) \leq R \}$$

The maximum possible neighborhood size is bounded:

$$|\mathcal{N}_R(p, t)| \leq N_{\max}(R) = |\{x \in \mathbb{Z}^3 : Q(x, \mathbf{0}) \leq R\}|$$

This is finite for all R , guaranteeing bounded computation per site per timestep. For $R = 25$, $N_{\max} \approx 523$.

3.3 Co-located Sites

If $Q(\text{pos}_p, \text{pos}_q) = 0$ for $p \neq q$, both sites are at the same lattice point. For gradient computation, co-located sites are included in the sum; the denominator $Q(p,q) = 0$ is replaced by 1 as a regularization convention.

4. Rational Geometry

4.1 Spread

For nonzero vectors $u, v \in \mathbb{Z}^3$:

$$s(u, v) = \frac{Q(u \times v, \mathbf{0})}{Q(u, \mathbf{0}) \cdot Q(v, \mathbf{0})}$$

where $u \times v$ is the integer cross product. This gives a rational number in $[0,1]$. Geometrically, $s = \sin^2(\theta)$ where θ is the angle between u and v .

4.2 Bifurcation Spread Constraint

For parent direction v_0 and child directions v_1, v_2 :

$$s(v_0, v_1) = s, \quad s(v_0, v_2) = s, \quad s(v_1, v_2) = 4s(1-s)$$

where $s \in \mathbb{Q} \cap (0,1)$ is the bifurcation spread parameter. The constraint $s(v_1, v_2) = 4s(1-s)$ follows from $\sin^2(2\theta) = 4\sin^2(\theta)\cos^2(\theta)$: the angle between children is twice the angle from parent to each child.

4.3 Constructing Child Directions

Given v_0 and s , pick a perpendicular vector w such that $s(v_0, w) = 1$ (i.e., $w \perp v_0$). Then set:

$$v_1 = v_0 + k \cdot w, \quad v_2 = v_0 - k \cdot w$$

where $k \in \mathbb{Q}$ is chosen so that $s(v_0, v_1) = s$. Solving:

$$k^2 = \frac{s}{1-s} \cdot \frac{Q(v_0, \mathbf{0})}{Q(w, \mathbf{0})}$$

If a rational k exists, use it. If not, scale v_0 and w by denominators until k is rational. This always terminates (multiply by denominators of $s/(1-s)$ and the quadrance ratio). The resulting v_1, v_2 are integer vectors satisfying the spread constraints exactly.

4.4 Direction Update (Bending)

For current direction $v_t \in \mathbb{Z}^3 \setminus \{0\}$, target direction $v_{\text{target}} \in \mathbb{Z}^3 \setminus \{0\}$, and bend strength $\alpha \in \mathbb{Z}_{\geq 0}$:

$$v_{t+1} = v_t + \alpha \cdot v_{\text{target}}$$

This is an integer vector. The default is $\alpha = 1$.

Position update on EXTEND:

$$\text{pos}_{\text{new}} = \text{pos}_p + v_{t+1}$$

5. Decision States

5.1 State Set

$$T_{\{p,t\}} \in \{+1, 0, -1, -2\}$$

Value Name Meaning

+1 EXTEND Grow or bifurcate

0 SENSE Adapt, redirect, remain

-1 RESTRICT Halt permanently, lose all energy

-2 CONSERVE Retract, return energy to parent

5.2 Gradient Estimation

For $p \in \mathcal{A}_t$:

$$\nabla \mathcal{E}(p,t) = \sum_{q \in \mathcal{N}_R(p,t)} \frac{E(q,t) - E(p,t)}{\max(Q(\text{pos}_p, \text{pos}_q), 1)} \cdot (\text{pos}_q - \text{pos}_p)$$

The trail gradient:

$$\nabla \text{Trail}(p,t) = \sum_{q \in \mathcal{N}_R(p,t)} \frac{\text{Trail}(\text{pos}_q) - \text{Trail}(\text{pos}_p)}{\max(Q(\text{pos}_p, \text{pos}_q), 1)} \cdot (\text{pos}_q - \text{pos}_p)$$

The effective gradient:

$$\nabla_{\text{eff}}(p,t) = \nabla \mathcal{E}(p,t) - \gamma \cdot \nabla \text{Trail}(p,t)$$

5.3 Gradient Magnitude

$$|\nabla_{\text{eff}}(p,t)|_Q = Q(|\nabla_{\text{eff}}(p,t)|, \mathbf{0})$$

5.4 Empty Neighborhood

If $\mathcal{N}_R(p,t) = \varnothing$:

$$\nabla_{\text{eff}}(p,t) = \mathbf{0}, \quad |\nabla_{\text{eff}}(p,t)|_Q = 0$$

5.5 Decision Function (Mutually Exclusive)

Conditions are evaluated in strict priority order.

Condition 1 — CONSERVE ($T = -2$)

$$(\text{parent}_p \neq \varnothing) \wedge (\text{parent}_p \in \mathcal{A}_t) \wedge \big(\text{pressure}(p,t) > P_{\text{hibernate}} \vee \text{external_hibernate_signal}(p,t) = \text{true} \big)$$

The hibernate pressure threshold defaults to $P_{\text{hibernate}} = 0.8 \cdot P_{\text{erode}} = 40$. The external_hibernate_signal is a domain-specific hook; defaults to false.

Condition 2 — RESTRICT ($T = -1$)

$$(E(p,t) < \tau) \vee \left(\min_{q \in \mathcal{N}_R(p,t)} Q(\text{pos}_p, \text{pos}_q) < \iota^2 \right)$$

But only if Condition 1 is false. The inequality $\min Q < \iota^2$ is strict. Exact equality does not trigger RESTRICT.

On RESTRICT: $E(p, t+1) = 0$, type_p becomes inert. The site permanently halts and all remaining energy is abandoned.

Condition 3 — EXTEND ($T = +1$)

$$(E(p,t) \geq \tau) \wedge \left(\min_{q \in \mathcal{N}_R(p,t)} Q(\text{pos}_p, \text{pos}_q) > \iota^2 \right) \wedge (|\nabla_{\text{eff}}(p,t)|_Q > \eta^2)$$

Additional constraint: If $\text{pressure}(p,t) \geq P_{\text{bifurcate}}$, require $E(p,t) \geq \epsilon_{\text{split}} + 2\tau$. This ensures both children have energy at least τ to survive one step.

If pressure triggers bifurcation but energy is insufficient, EXTEND is false; the site falls through to Condition 4.

Condition 4 — SENSE ($T = 0$)

All remaining cases.

Within SENSE, the site may adapt its behavior:

- If $\iota^2 < \min Q \leq (\iota + \delta)^2$: the site is in the approach zone. It may reduce its direction magnitude and/or adjust direction away from the nearest neighbor.
- If $|\nabla_{\text{eff}}|_Q \leq \eta^2$: the gradient is flat. The site may perturb its direction to explore the local energy landscape.
- Otherwise: the site maintains its current direction but does not extend.

SENSE-to-CONSERVE fallback: If $T = 0$ and $E(p,t) - \epsilon_{\text{sense}} < \tau$, and $\text{parent}_p \neq \text{nothing}$ and $\text{parent}_p \in \mathcal{A}_t$, then $T = -2$ (CONSERVE). This prevents sites from wasting their last energy on a SENSE that would trigger RESTRICT next step. The fallback applies in the narrow window $\tau \leq E < \tau + \epsilon_{\text{sense}}$.

5.6 Decision Function Parameters

Symbol Meaning Domain

τ Energy threshold $\mathbb{Q}_{\geq 0}$

ι Contact inhibition distance $\mathbb{Q}_{>0}$

η^2 Squared gradient threshold $\mathbb{Q}_{\geq 0}$

δ Sense boundary width $\mathbb{Q}_{\geq 0}$

γ Trail avoidance weight $\mathbb{Q}_{\geq 0}$

6. Energy Dynamics

6.1 Energy Cost Function

$$\kappa(T) = \begin{cases} \epsilon_{\text{extend}}, & T = +1 \\ \epsilon_{\text{sense}}, & T = 0 \\ E(p,t), & T = -1 \\ \epsilon_{\text{conserve}}, & T = -2 \end{cases}$$

All costs are in $\mathbb{Q}_{\geq 0}$. On RESTRICT ($T = -1$), the cost is the site's entire remaining energy; the site is set to $E = 0$.

Critical requirement for eventual inactivity: $\epsilon_{\text{extend}} > 0$ and $\epsilon_{\text{sense}} > 0$. Define:

$$\epsilon_{\min} = \min(\epsilon_{\text{extend}}, \epsilon_{\text{sense}}) > 0$$

6.2 Standard Energy Update

For $T_{\{p,t\}} \in \{+1, 0, -1\}$:

$$E(p, t+1) = E(p, t) - \kappa(T_{\{p,t\}})$$

For $T = -1$ (RESTRICT), this becomes $E(p, t+1) = 0$.

6.3 CONSERVE Update

For $T_{\{p,t\}} = -2$, let $q = \text{parent}_p$:

Phase 2 of the update processes CONSERVE sites in topological order (tips toward root). A temporary energy buffer $E_{\{\text{buffer}\}}$ is used:

$$E_{\{\text{buffer}\}}(q) = E(q, t) + \beta \cdot (E(p, t) - \varepsilon_{\{\text{conserve}\}})$$

$$E(p, t+1) = 0$$

$$p \notin \mathcal{A}_{t+1}$$

$$\text{pressure}_{\{\text{buffer}\}}(q) = \text{pressure}(q, t) + \beta \cdot \text{pressure}(p, t)$$

where $\beta \in (0, 1)$. The strict inequality $\beta < 1$ ensures net energy loss.

Phase 5 applies costs using buffered values. If multiple children CONSERVE to the same parent, buffer accumulations are additive. A parent that itself CONSERVEs uses its buffered energy for its own transfer. Since topological ordering processes children before parents, a parent's CONSERVE always includes any child contributions.

Pressure transfers are scalar. The receiving parent applies transferred pressure to its own local context, not the child's. A child blocked by a stone at position X passes frustration magnitude to the parent, which applies it to whatever obstacle the parent faces at its own position.

6.4 Bifurcation Energy Split

Parent site p splits into children c_1, c_2 . Parent energy becomes 0:

$$E(c_1, t+1) = E(c_2, t+1) = \frac{E(p, t) - \varepsilon_{\{\text{split}\}}}{2}$$

Requires $E(p, t) \geq \varepsilon_{\{\text{split}\}}$. Children inherit $\text{parent} = p$.

If gradient bias is used:

$$E_{\{\text{closer}\}} = \frac{E(p,t) - \varepsilon_{\{\text{split}\}}}{2} + \Delta E_{\{\text{bias}\}}$$

$$E_{\{\text{farther}\}} = \frac{E(p,t) - \varepsilon_{\{\text{split}\}}}{2} - \Delta E_{\{\text{bias}\}}$$

6.4a Single Extension Energy

When a site EXTENDs without bifurcation (single child), parent energy becomes 0:

$$E_{\{\text{child}\}} = E_{\{\text{parent}\}} - \varepsilon_{\{\text{extend}\}} - \varepsilon_{\{\text{spawn}\}}$$

where $\epsilon_{\text{spawn}} \in \mathbb{Q}_{\geq 0}$ is the overhead for creating any child. Default: $\epsilon_{\text{spawn}} = \epsilon_{\text{split}} / 2$.

6.5 Conflict Loser Energy

All energy costs are applied in Phase 5 based on each site's final decision after conflict resolution. A site that originally chose EXTEND in Phase 1 but was forced to SENSE by a conflict pays only the SENSE cost:

$$E(\text{loser}, t+1) = E(\text{loser}, t) - \epsilon_{\text{sense}}$$

6.6 Open-System Extension

$$E(p, t+1) = E(p, t) - \kappa(T_{\{p,t\}}) + I(p, t) + \sum_{q \in \mathcal{N}_R(p,t)} J_{\{q \rightarrow p, t\}}$$

$I(p,t)$ is external input. $J_{\{q \rightarrow p, t\}}$ is local transfer satisfying $J_{\{q \rightarrow p, t\}} = -J_{\{p \rightarrow q, t\}}$. The eventuality theorem applies only when $I(p,t) = 0$ for all p, t .

7. Pressure Dynamics

7.1 Pressure Accumulation

Pressure accumulates from two sources, both contributing to a single accumulator:

Obstacle contact: When an EXTEND candidate is rejected due to $\text{Env}(x) = \text{obstacle}$:

$$\text{pressure}(p, t+1) = \text{pressure}(p, t) + \Delta P_{\text{contact}}$$

Conflict loss: When an EXTEND candidate loses a spatial conflict to another site:

$$\text{pressure}(\text{loser}, t+1) = \text{pressure}(\text{loser}, t) + \Delta P_{\text{conflict}}$$

The site does not distinguish between obstacle pressure and competition pressure.

On RESTRICT, pressure resets to 0. The site's accumulated constraint history is discarded—it has given up entirely. On CONSERVE, pressure transfers to the parent (Section 6.3). This asymmetry is intentional: CONSERVE preserves experiential information; RESTRICT abandons it.

7.2 Pressure-Driven Bifurcation

If $\text{pressure}(p, t) \geq P_{\text{bifurcate}}$ and $T_{\{p,t\}} = +1$, bifurcation is triggered with spread:

$$s = s_{\min} + (s_{\max} - s_{\min}) \cdot \min\left(1, \frac{\text{pressure}(p,t)}{P_{\text{erode}}}\right)$$

7.3 Pressure-Driven Erosion

If $\text{pressure}(p, t) \geq P_{\text{erode}}$:

$$\text{Env}(x) \rightarrow \text{empty} \quad \text{where } x \text{ is the contact position}$$

$\text{pressure}(p, t+1) = 0$

Erosion occurs at the contact position. The parent's next EXTEND attempt at that position will succeed.

7.4 Pressure Parameters

Symbol Meaning Default

$\Delta P_{\text{contact}}$ Pressure per obstacle hit 5

$\Delta P_{\text{conflict}}$ Pressure per lost conflict 2

$P_{\text{bifurcate}}$ Threshold for branching 20

P_{erode} Threshold for erosion 50

$P_{\text{hibernate}}$ Threshold for CONSERVE 40

$s_{\min}$ Minimum spread 1/16

$s_{\max}$ Maximum spread 1/2

8. Update Execution Sequence

The update proceeds in ordered phases.

Phase 1: Decision Computation

For all $p \in \mathcal{A}_t$, compute $T_{p,t}$ using the decision function (Section 5.5) evaluated on S_t .

Phase 2: CONSERVE Processing

Process sites with $T_{p,t} = -2$ in topological order from tips toward root:

- A site p may CONSERVE only if no child of p that also chose CONSERVE is still unprocessed.
- Transfer energy and pressure to parent_p via the buffer mechanism (Section 6.3).
- Mark p for removal from $\mathcal{A}_{t+1}$.

Phase 3: Candidate Generation

For all p with $T_{p,t} = +1$, compute candidate new positions:

- Extension: candidate position = $\text{pos}_p + \text{dir}_p$. New site type = tip; p becomes type branch.
- Bifurcation: two candidate positions = $\text{pos}_p + v_1, \text{pos}_p + v_2$, with v_1, v_2 satisfying spread constraints (Section 4.3). Two new tip sites; p becomes type branch.

Parent energy becomes 0 in both cases.

Phase 4a: Obstacle Rejection

For each candidate position x :

- If $\text{Env}(x) = \text{obstacle}$: reject the candidate. Mark the parent as "obstacle-blocked."
- If $\text{Env}(x) = \text{target}$: candidate succeeds. Mark parent as "target-reached."

Obstacle-blocked parents: Remain EXTEND for this timestep, pay ϵ_{extend} , accumulate $\Delta P_{\text{contact}}$, do not move. Children are not spawned.

Phase 4b: Site-Site Conflict Resolution

For all remaining candidate positions within quadrance ι^2 of each other or within ι^2 of an existing site position:

- The candidate with lower parent energy yields.
- If equal energy, lower parent site identifier yields.
- Yielding candidate: parent switches to SENSE, pays $\varepsilon_{\text{sense}}$, accumulates $\Delta P_{\text{conflict}}$.
- Winning candidate: proceeds to Phase 5.

Phase 5: State Application

- Apply energy costs for all sites according to their final decisions, using buffered values for sites that received CONSERVE transfers.
- Add successfully spawned children to $\mathcal{A}_{t+1}$.
- Update positions and directions for extended/bifurcated parents.
- Remove CONSERVE sites from $\mathcal{A}_{t+1}$.
- Sites with $T = -1$ remain in $\mathcal{A}_{t+1}$ with type inert and zero energy.

Phase 6: Trail and Pressure Updates

- Apply trail deposition at all active (non-inert) positions in $\mathcal{A}_{t+1}$.
- Apply trail decay globally.
- Apply pressure updates.

9. Type System

9.1 Type Set

$\text{Type} = \{\text{seed}, \text{tip}, \text{branch}, \text{sensor}, \text{inert}\}$

9.2 Type Transition Function

Current type EXTEND (+1) SENSE (0) RESTRICT (-1) CONSERVE (-2)
seed branch sensor inert removed
tip branch sensor inert removed
branch branch sensor inert removed
sensor tip sensor inert removed
inert inert inert inert removed

Note: After EXTEND, parent type becomes branch with $E = 0$. Branch sites do not spawn additional children in the base framework. Branch reactivation (lateral tip spawning) is an open extension (Open Question #3).

9.3 Spawn Rules

Parent action	Children spawned	Child types
EXTEND (tip, sensor)	1 child	tip
EXTEND (seed)	1 child	tip
EXTEND + bifurcation	2 children	tip, tip
EXTEND (branch)	0 children	—

9.4 Site Fate on Termination

Termination Mode Type Change In $\mathcal{A}_{t+1}$? Energy
RESTRICT becomes inert yes zero
CONSERVE deleted no N/A

10. Environment

10.1 Environment Map

Env : $\mathbb{Z}^3$ to $\{\text{empty}, \text{obstacle}, \text{target}\}$

Initialized before simulation. Modified only by pressure-driven erosion (Section 7.3).

10.2 Obstacle Contact

Agents do not have direct access to Env. They sense obstacles only through contact during an EXTEND attempt, as specified in Phase 4a (Section 8).

11. Environmental Trails

11.1 Trail Map

Trail : $\mathbb{Z}^3$ to $\mathbb{Q}_{\geq 0}$

Initialized to zero. Stored as a sparse map in implementation. Support is finite at all t.

11.2 Deposition

For each p in $\mathcal{A}_{t+1}$ with $\text{type}_p \neq \text{inert}$:

$\text{Trail}(\text{pos}_p) \leftarrow \text{Trail}(\text{pos}_p) + \Delta_{\text{trail}}$

Inert sites do not deposit trails. They are structural remnants, not active agents.

11.3 Decay

$\text{Trail}(x) \leftarrow \max(0, \text{Trail}(x) - \Delta_{\text{decay}}) \quad \forall x \in \mathbb{Z}^3$

In practice, decay is applied only to the finite support set of Trail.

11.4 Parameters

Symbol	Meaning	Default
Δ_{trail}	Trail per visit	0.5

\Delta_{\text{decay}} Decay per timestep 0.01
\gamma Avoidance weight 1.0

12. Master Update (Two-Phase Form)

Phase A — Decision:

$$T_{\{p,t\}} = \Theta(x_{\{p,t\}}, E(p,t), \mathcal{N}_R(p,t), \mathcal{P}) \quad \forall p \in \mathcal{A}_t$$

Phase B — Execution:

$$S_{\{t+1\}} = \Psi(S_t, \{T_{\{p,t\}}\}_{p \in \mathcal{A}_t})$$

where Ψ encompasses Phases 2–6 from Section 8.

13. Eventual Inactivity Theorem

Statement

Let a Natural Math process satisfy:

- 1. $E_{\text{tot}}(0) < \infty$.
- 2. $l(p,t) = 0$ for all p, t (no external energy).
- 3. $\epsilon_{\text{extend}} > 0$ and $\epsilon_{\text{sense}} > 0$.
- 4. $\beta \in (0,1)$ (CONSERVE loses energy).
- 5. Energy transfers $J_{q \rightarrow p}$ are conservative.

Then there exists $T < \infty$ such that for all $t \geq T$ and all $p \in \mathcal{A}_t$:

$$T_{\{p,t\}} \in \{-1\} \text{ or } p \text{ is inert}$$

The system reaches a frozen state: no further EXTEND, SENSE, or CONSERVE decisions occur. RESTRICT and inert sites may remain, but the active process has halted.

Proof

Define $E_{\text{tot}}(t) = \sum_{p \in \mathcal{A}_t} E(p, t)$.

EXTEND steps: Each consumes $\epsilon_{\text{extend}} \geq \epsilon_{\min} > 0$.

SENSE steps: Each consumes $\epsilon_{\text{sense}} \geq \epsilon_{\min} > 0$.

CONSERVE steps: Net change $\Delta E = -(1-\beta)E_{\text{child}} - \beta\kappa_{\text{conserve}} < 0$ since $\beta < 1$ and $E_{\text{child}} \geq 0$.

RESTRICT steps: $E(p, t+1) = 0$. Energy decreases by $E(p,t) \geq 0$. These steps are not counted in the active timestep bound.

Transfers J: Sum to zero, no effect on E_{tot} .

Conflicts: Losers pay $\epsilon_{\text{sense}} \geq \epsilon_{\min}$.

In any timestep where at least one EXTEND, SENSE, or CONSERVE occurs, or any conflict produces a SENSE, total energy decreases by at least $\epsilon_{\min}$. RESTRICT-only timesteps do not count; they represent the system entering its frozen state. Invariants I2 and I3 ensure inert and RESTRICT sites hold no energy that could fuel further activity.

Let N_{active} be the number of timesteps with at least one such energy-decreasing event. Since $E_{\text{tot}}(t) \geq 0$ and $E_{\text{tot}}(0)$ is finite:

$$N_{\text{active}} \leq \left\lfloor \frac{E_{\text{tot}}(0)}{\epsilon_{\min}} \right\rfloor$$

After at most this many active timesteps, no EXTEND, SENSE, or CONSERVE can occur. All remaining sites must have $T = -1$ or be inert. The system is frozen. $\square$

14. Field Formulation

$$\mathfrak{F}: \mathbb{Z}^3 \times \mathbb{N} \rightarrow \{\text{active}, \text{inert}, \text{empty}\} \times \mathbb{Q}_{\geq 0} \times \text{Type} \times \mathbb{Z}^3$$

The update:

$$\mathfrak{F}(\cdot, t+1) = \begin{cases} \mathfrak{F}(\cdot, t), & \text{if all sites at } t \text{ are inert or empty} \\ \Psi(\mathfrak{F}(\cdot, t)), & \text{otherwise} \end{cases}$$

15. Open Questions

The following must be specified per application:

1. External hibernate signal: What domain-specific conditions trigger CONSERVE beyond pressure? (Seasonal cues, policy changes, resource shocks.)
 2. Bifurcation trigger beyond pressure: Does energy surplus alone trigger branching? Default: no.
 3. Branch spawning: When does a branch site spawn a lateral tip? Default: never. Note that after Version 2.2, branches have $E = 0$ after spawn; any branch reactivation would require external energy input or a redistribution mechanism.
 4. Trail decay rate: Domain-dependent.
 5. Obstacle representation: How are obstacles stored and queried?
 6. Gradient bias ΔE_{bias} : Default is 0.
 7. Exploration drive: Optional n_{patience} parameter forces EXTEND after n consecutive SENSE steps. Default: ∞ (disabled).
 8. Reproduction: The framework currently models a single acyclic growth structure—one forest with one or more seed origins, growing until it freezes. Reproduction would allow a structure to spawn independent seeds, creating a population of forests across time.
- Design constraint: Any reproduction mechanism must obey the same principles as the rest of the framework—local information only, finite energy cost, no global knowledge, emergent rather than goal-directed.
- Proposed entry condition: A tip becomes eligible for reproduction when three conditions hold simultaneously: sufficient energy exists, local growth efficiency has fallen, and the site is healthy enough to invest resources in dispersal.
- Proposed metric: Expansion Efficiency. Define for a tip p :
- $$\text{ExpansionEfficiency}(p, t) = \frac{\text{NovelPositions}(p, t-w, t)}{\text{EnergySpent}(p, t-w, t)}$$
- where NovelPositions is the count of distinct lattice positions first visited by p or its recent descendants within the window $[t-w, t]$, and EnergySpent is the total energy consumed by p and its recent descendants in that window. Both are locally computable from the site's subtree state.
- Proposed decision state: A fifth decision value, $T = -3$ (REPRODUCE), inserted at priority 2.5 (after RESTRICT, before EXTEND):
- $$(E(p, t) \geq E_{\text{reproduce}}) \wedge (\text{ExpansionEfficiency}(p, t) < \eta_{\text{reproduce}}) \wedge (\text{parent}_p \neq \text{nothing})$$
- Action: Compute dispersal position $\text{pos}_{\text{seed}} = \text{pos}_p + \mathbf{d}$ where $\mathbf{d}$ is a dispersal vector (domain-specific: random, gradient-following, pressure-directed, or parent-directed). Create a new seed site with energy $E_{\text{seed}} = \min(E(p, t) - \epsilon_{\text{reproduce}}, E_{\text{max_seed}})$ and inherited (or mutated) parameters. Parent pays $E_{\text{seed}} + \epsilon_{\text{reproduce}}$ and becomes inert.

Lifecycle extension: REPRODUCE adds a second terminal state—completion. The reproducing site dies, but its lineage continues in a new, independent forest.

Theorem impact: Each individual forest still obeys the eventuality theorem. Population persistence depends on whether the expected energy return from reproduction exceeds the energy invested. If reproduction is dissipative on average, the total population eventually freezes. If external energy is available, populations can persist indefinitely.

Priority note: This is specified as an extension point, not a core feature. Version 2.2 models individual growth structures completely. Reproduction is the natural bridge to population-level dynamics, reserved for a future version once the growth simulator is validated.

Part 3: Design Rationale

Why Synchronous Evaluation?

All sites evaluate their decisions based on the same pre-update state S_t . Alternatives considered:

- Asynchronous (random order): Outcome depends on update order. Two runs with identical initial state can diverge. Termination proof becomes order-dependent.
- Continuous time: Requires differential equations; loses discrete computational guarantee.

Synchronous evaluation guarantees consistent information. The two-phase design eliminates circular dependencies. All Phase 1 decisions are independent and parallelizable.

Why Topological Ordering for CONSERVE?

When multiple sites retract, a child might CONSERVE to its parent, and that parent might also CONSERVE. Processing children before parents ensures the parent receives the child's energy before deciding whether to retract. This matches biological intuition—retraction waves move from tips toward the base.

Why Does SENSE Cost Energy?

Three reasons:

1. Physical: Maintenance costs energy even when not growing. A root tip is metabolically active. A company's R&D department burns budget while exploring options.
2. Mathematical: If SENSE were free, the termination proof would fail—a system could SENSE forever. Making SENSE cost energy is the simplest way to guarantee finite behavior without additional assumptions.
3. Design: Positive SENSE cost creates a meaningful tradeoff. Extended sensing has opportunity cost—energy spent sensing is not spent extending. This forces balance between exploration and exploitation.

Why the 5:1 Cost Ratio?

The defaults ($\epsilon_{\text{extend}} = 10$, $\epsilon_{\text{sense}} = 2$) encode three strategies at an obstacle:

- Dither (SENSE): Cheap (2/step) but ineffective—never moves. Slow death at permanent obstacles.
- Push through (repeated EXTEND): Expensive (10/attempt, 40 energy to reach $P_{\text{bifurcate}} = 20$, plus bifurcation overhead ~55–60 total). Effective—triggers branching or erosion.
- Retreat (CONSERVE): Cheap (1 plus recovery loss). Abandons the direction; reallocates energy to alternatives.

No central planner chooses. Local energy and pressure determine the outcome.

Why RESTRICT Abandons All Energy?

On RESTRICT, a site's energy is set to zero—it is not returned to the parent. This is in contrast to CONSERVE, which returns a β fraction. The asymmetry is intentional:

- RESTRICT: "I am dead. Nothing to pass on." The site hit a hard limit (energy depletion or crowding) and perished in place. Its remaining energy is lost to the environment.

· CONSERVE: "I am retreating strategically. Here is what I have left." The site recognized unfavorable conditions and pulled back, preserving resources for the rest of the structure.

This preserves Invariant I2 (inert sites have zero energy) and creates a meaningful strategic choice: retreat early to preserve resources, or push until the end and risk losing everything.

Why Scalar Pressure Cascades?

Pressure transfers from child to parent on CONSERVE carry magnitude only—not directional information about the source of blockage. A child blocked by a stone at position X passes frustration magnitude to the parent, which applies it to whatever obstacle the parent faces at its own position.

This is a deliberate simplification. It models "this branch is struggling" as a signal, without requiring the parent to know why or where. In biological terms: a root tip encountering a stone sends a biochemical signal upward; the parent root responds by trying harder in its own direction, not by targeting the child's stone.

Why the SENSE-to-CONSERVE Fallback?

A site with energy in the narrow window $\tau \leq E < \tau + \epsilon_{\text{sense}}$ that chooses SENSE would pay its SENSE cost, drop below τ , and RESTRICT on the next timestep—losing all remaining energy. The fallback intercepts this: if SENSE would push energy below τ , the site CONSERVEs instead, returning a β fraction to the parent.

The window is small (default: $8 \leq E < 10$), but the energy preserved is meaningful relative to the total budget. The fallback encodes a simple heuristic: "If one more step of waiting would kill me, retreat now."

Why Parent Acyclicity is an Invariant

CONSERVE requires a unique parent. Topological ordering requires a partial order. Cycles would allow energy to circulate forever, violating the eventuality theorem. The invariant holds because children are only created by EXTEND from existing parents, and parents are never created by children.

Why Integer Lattice?

$\mathbb{Z}^3$ is locally finite by construction. This gives bounded neighborhood size ($N_{\max}$), exact integer position updates, and guaranteed finite computation per site per timestep. The cost is losing continuous direction resolution, but spread-based angle measurement on integer vectors preserves exact rational geometry.

Part 4: Worked Examples

Example 1: Root Growth Through Soil

Domain Mapping

- Framework Concept Root Domain
- Agent Root tip (apical meristem)
- Energy Stored carbohydrate
- Obstacle Stone, compacted soil
- Trail Depleted nutrient zone
- Pressure Mechanical resistance
- CONSERVE Seasonal dieback, stress abortion

Walkthrough

Timestep 1: Seed at (0,0,0), $E = 400$, direction (0,1,0), type = seed. Neighborhood empty. $|\nabla_{\text{eff}}|_Q = 0 \leq \eta^2$. Decision: SENSE. Cost: 2. Energy: 398.

Timestep 2: Same. Decision: SENSE. Cost: 2. Energy: 396.

Timestep 3: Target tropism provides gradient. Decision: EXTEND. Seed spawns child tip at (0,1,0). Seed becomes branch. Child energy: $396 - 10 - 7 = 379$.

Timestep 4: Tip at (0,1,0), $E = 379$. EXTEND to (0,2,0). New child energy: $379 - 17 = 362$.

Timestep 5: Tip at (0,2,0), $E = 362$. Stone at (0,3,0). Candidate rejected (Phase 4a). Parent remains EXTEND. Pays 10. Energy: 352. Pressure: 5.

Timestep 6: Tip at (0,2,0), $E = 352$, pressure = 5. $\min Q$ to parent = $1 = \iota^2$ (boundary case). Conditions 2 and 3 both fail. SENSE. Cost: 2. Energy: 350. Direction bent slightly.

Timestep 7: Tip at (0,2,0), $E = 350$, pressure = 5. EXTEND attempt offset from vertical. Still hits stone. Rejected. Pays 10. Energy: 340. Pressure: 10.

Timesteps 8–9: Repeated contact. Pressure reaches 15, then 20.

Timestep 10: Pressure = 20. $P \geq P_{\text{bifurcate}}$. EXTEND, bifurcation triggered. Two children with spread $s = 1/4$ ($\sim 60^\circ$), biased toward target. Children energy: $(340 - 14)/2 = 163$ each. Parent becomes branch.

Total bypass cost: 4 EXTEND attempts (40) + bifurcation overhead (14) + 2 SENSE steps (4) = 58 energy.

Example 2: Urban Development (Sketch)

Framework Concept Urban Domain
Agent Development zone
Energy Available capital
Obstacle Zoning restriction, protected land
Trail Market saturation
Pressure Land value pressure, density
CONSERVE De-zoning, land banking

Growth extends along transportation corridors. Zoning boundaries create obstacles. Pressure builds at restricted edges, triggering either policy change (erosion) or leapfrog development (bifurcation). Trail represents market saturation—recently developed areas have absorbed demand, steering growth toward underserved territory. Trail decay models demographic turnover, allowing eventual redevelopment.

Example 3: Fungal Mycelium (Sketch)

Framework Concept Fungal Domain
Agent Hyphal tip
Energy Internal nutrient reserves
Obstacle Air gap, toxic substrate
Trail Depleted substrate, self-recognition
Pressure Mechanical resistance
CONSERVE Hyphal retraction, resource reallocation

Hyphae EXTEND into nutrient-rich areas, SENSE at boundaries, RESTRICT when exhausted, and CONSERVE to retract exploratory hyphae. Trail avoidance (self-recognition) prevents self-crossing. Network fractal dimension emerges from branching spread and nutrient distribution.

Parameter Groups

Energy Budget

Parameter Meaning Default
 τ Minimum energy to extend 8
 ϵ_{extend} EXTEND cost 10
 ϵ_{sense} SENSE cost 2
 $\epsilon_{\text{restrict}}$ RESTRICT cost all remaining
 $\epsilon_{\text{conserve}}$ CONSERVE cost 1
 ϵ_{split} Bifurcation overhead 14
 ϵ_{spawn} Overhead per child 7
 β CONSERVE recovery fraction 0.8

Geometry

Parameter Meaning Default
R Neighborhood quadrance bound 25
 ι Minimum spacing 1
 α Bend strength 1
s Bifurcation spread 1/4

Pressure

Parameter Meaning Default
 $\Delta P_{\text{contact}}$ Pressure per obstacle hit 5
 $\Delta P_{\text{conflict}}$ Pressure per lost conflict 2
 $P_{\text{bifurcate}}$ Threshold for branching 20
 P_{erode} Threshold for erosion 50
 $P_{\text{hibernate}}$ Threshold for CONSERVE 40
 s_{min} , s_{max} Spread range under pressure 1/16, 1/2

Memory

Parameter Meaning Default
 Δ_{trail} Trail per visit 0.5
 Δ_{decay} Decay per timestep 0.01
 γ Avoidance weight 1.0

Gradient

Parameter Meaning Default
 η^2 Squared gradient threshold 1
 δ Sense boundary width 2

Optional Domain Parameters

Parameter Meaning Default

ΔE_{bias} Gradient-aligned child bonus 0
 n_{patience} Max SENSE before forced EXTEND ∞

Scope and Status

Natural Math is a formal modeling language for acyclic spatial growth processes under finite constraints. Its strongest result is the eventual inactivity theorem for closed systems. Its value lies in providing a common, testable structure—the four decision states, local agency, energy accounting, and lifecycle—applicable across domains that share the four defining properties.

Version 2.2 is internally consistent and implementable. The open questions identify extension points, not gaps. Reproduction (Open Question #8) is the natural next step: from modeling an individual growth structure to modeling a population of structures across time. That step awaits empirical validation of the core framework.

The next stage is implementation and testing against observed morphologies in root growth, fungal networks, and vascular systems. The framework is ready for that work.